

Trainings ▾

Preparatory Steps

Note : Removal of the sensor tubes does **not** require sensor removal. Sensor removal is necessary **only** if the sensor adapter is being replaced.

- Remove the exhaust gas pressure sensor if the adapter is being replaced. Refer to Procedure 019-376 in Section 19. (</qs3/pubsys2/xml/en/procedures/88/88-019-376.html>) Reference the Troubleshooting and Repair Manual, CM870 Electronic Control System, ISM Engines, Bulletin 4021381.



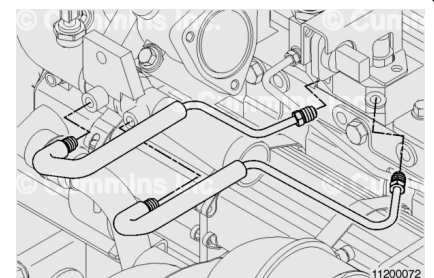
Remove

Automotive with CM875, CM870 and CM570

Loosen the tube nuts at the EGR connection tube and the sensor adapter.

Do **not** distort the sensor tubes.

Remove the EGR differential pressure sensor tubes.

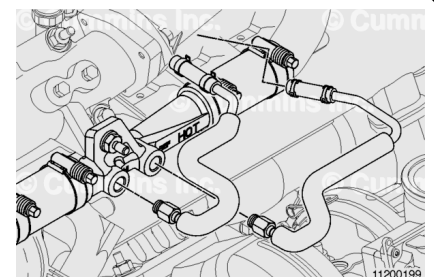


Automotive with CM876

Loosen the tube nuts at the venturi connections.

Loosen the hose clamps at the differential pressure sensor adapter.

Remove the EGR differential pressure sensor tubes.

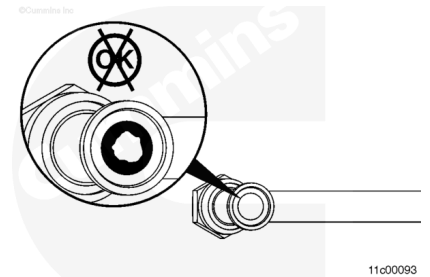


Clean and Inspect for Reuse

Automotive with CM875, CM870 and CM570

Carbon deposits inside the tubes can cause a yellow check engine light, plus any combination of the following:

- Low power
- Black smoke
- Engine surge
- Engine runs rough
- Exhaust gas recirculation valve delta pressure fault code.



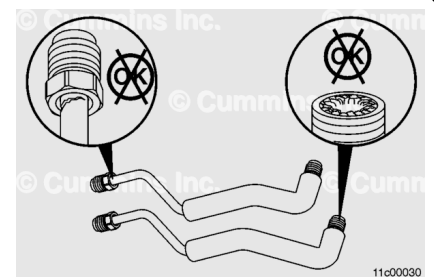
Note : If the EGR cooler is being replaced due to damage that allowed coolant to leak into the EGR differential pressure sensor tubes, the tubes must be replaced.

Check the inside of the tubes.

If a tube is clogged or partially clogged, replace the tube.

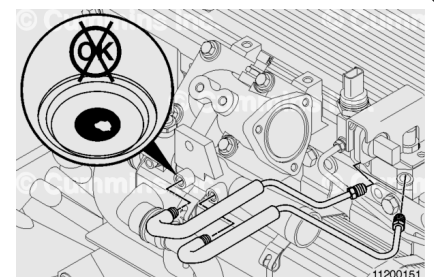
Check the tubes for cracks or thread damage.

Replace the tubes if they are cracked or the threads are damaged.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect the mating orifices for the EGR crossover tubes.

If clogged, saturate the carbon deposit with a mineral based solvent, or equivalent.

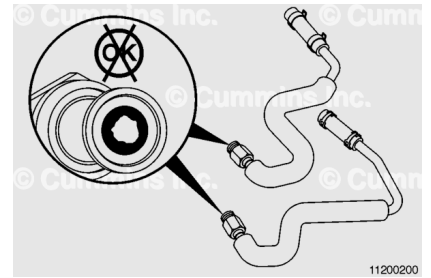
Carefully clean the debris from the mouth of the orifice, making sure **not** to damage the orifice.

Dry with compressed air.

Automotive with CM876

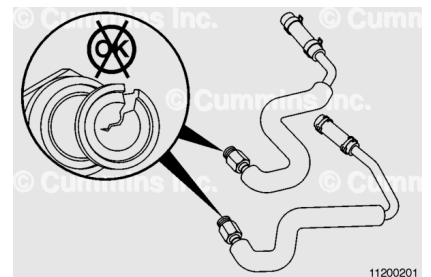
Carbon deposits inside the tubes can cause a yellow check engine light, plus any combination of the following:

- Low power
- Engine surge
- Engine runs rough
- Exhaust gas recirculation valve delta pressure fault code
- Progressive damage to the Cummins® particulate filter.



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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Check the inside of the tubes.

If a tube is partially clogged, saturate the inside of the tube with a mineral based solvent, or equivalent.

Carefully clean the debris from the mouths of the tubes. Make sure **not** to damage the tube.

If a tube is fully clogged or otherwise unable to be cleaned, replace the tube.

Clean the tube with safety solvent.

Dry with compressed air.

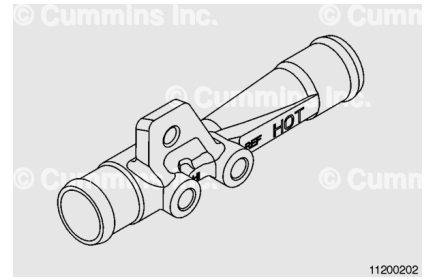
Check the tubes, hose, and hose clamps for cracks or thread damage.

Replace the tubes if the tubes or hose is damaged.

Replace the hose clamps if damaged.

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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect the mating orifices for the EGR crossover tubes.

If clogged, saturate the carbon deposit with a mineral based solvent, or equivalent.

Carefully clean the debris from the mouth of the orifice, making sure **not** to damage the orifice.

Dry with compressed air.

Clean and inspect the venturi. Refer to Procedure 011-025 in Section 11. (</qs3/pubsys2/xml/en/procedures/35/35-011-025-tr.html>)

Install

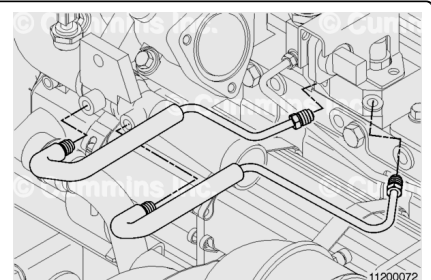
Automotive with CM875, CM870 and CM570

Loosen the EGR differential pressure sensor adapter mounting capscrews, but do **not** remove. This is so the tubes are **not** distorted during installation.

Make sure **not** to cross-connect tubes or a fault code will result.

Apply a light lubricant (CRC 3-36, WD-40, or equivalent) to the tube threads and install the tubes. The lubricant **must** be applied inside the tube nut, directing the spray on the threads and around the sleeve and flange area.

Tighten the tube nuts.



Torque Value: 16 n•m [142 in-lb]

Position the insulation on the differential pressure sensor tubes so the tube nuts at the venturi connection are covered.

The insulation on the pressure tubes is designed to **only** cover the bottom portion of the tube which threads into the EGR transfer tube. This location proves to be most effective and prevents heat soaking of the EGR differential pressure sensor.

Tighten the EGR differential pressure sensor adapter mounting capscrews. Refer to Procedure 011-028 in Section 11.

(/qs3/pubsys2/xml/en/procedures/35/35-011-028-tr.html)

Automotive with CM876

Make sure **not** to cross-connect tubes or a fault code will result.

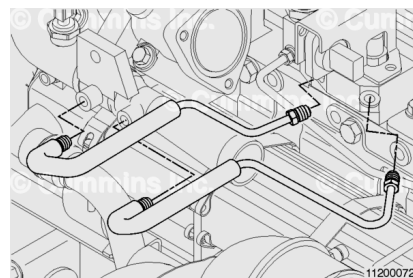
Apply a light lubricant (CRC 3-36, WD-40, or equivalent) to the tube threads and install the tubes. The lubricant **must** be applied inside the tube nut, directing the spray on the threads and around the sleeve and flange area

Tighten the tube nuts at the venturi connections.

Torque Value: 26 n·m [230 in-lb]

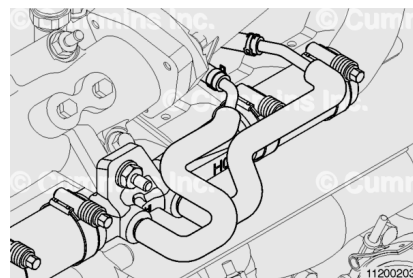
Tighten the hose clamps at the differential pressure sensor adapter.

Torque Value: 1.5 n·m [13 in-lb]



Position the insulation on the differential pressure sensor tubes so that the tube nuts at the venturi connection are covered.

The insulation on the pressure tubes is designed to cover **only** the bottom portion of the tube which threads into the EGR transfer tube. This is the most effective location and reduces the possibility of heat soaking of the EGR differential pressure sensor.

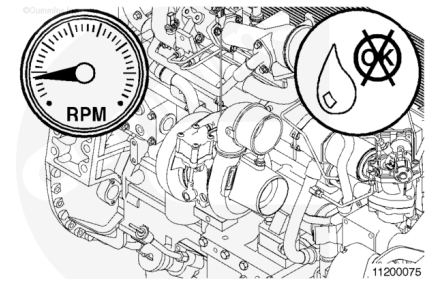


Finishing Steps

- If removed, install the exhaust gas pressure sensor. Refer to Procedure 019-376 in Section 19.
(/qs3/pubsys2/xml/en/procedures/88/88-019-376.html)

Reference the Troubleshooting and Repair Manual, CM870 Electronic Control System, ISM Engines, Bulletin 4021381.

- Start the engine and verify proper operation. Reference the Troubleshooting and Repair Manual, CM870 Electronic Control System, ISM Engines, Bulletin 4021381.
- Check for fault codes and leaks.



If damage resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013.html>)